

BudgetWise – Personal Finance and Spending Analysis

Python Project Report

1. Project Overview and Objective

This project involves cleaning, transforming, and analysing personal financial transaction data using Python/Excel and creating interactive Power BI dashboards to derive meaningful financial insights.

Objective: To analyse personal financial transactions and identify spending patterns, major expenses, payment behaviour, and areas where better budgeting can help manage money effectively.

2. Data Sources

Source Description: kaggle / BudgetWise Finance Data Analysis Dataset / <https://www.kaggle.com/datasets/kennathalexanderroy/budget-vs-actual-financial-dataset>

Timeline: Data collected during 2021 to 2024

Domain: Finance

3. Problem Statement

- Analyse personal spending patterns across categories and transaction types.
- Identify major spending categories and high-value transactions.
- Understand payment-mode behaviour.
- Examine spending trends over time.
- Identify areas where better budgeting and money management may be useful.

4. Attribute (Column / Features) Details

Attribute Name	Data Type	Description
Date	Date	Transaction date
Category	Categorical	Spending category
Amount	Numeric	Transaction amount
Payment Mode	Categorical	Payment method used
Location	Categorical	Transaction location
Transaction Type	Categorical	Income or expense type
Notes	Text	Additional transaction notes

5. Tools & Technologies

- **Python:** Data cleaning, preprocessing, EDA and visualization.
- **Excel:** Data review, cleaning and transformation where required.
- **Power BI:** Data modelling, DAX calculations, visualization, slicers and interactive dashboard creation.

6. Data Pre-Processing

- Checked and handled missing values.
- Standardized category, payment mode and location values.
- Cleaned the amount field and converted it to numeric format.
- Converted date to a proper date data type.
- Checked duplicate records.
- Cleaned unwanted or inconsistent notes values.
- Validated the final dataset before Power BI import.

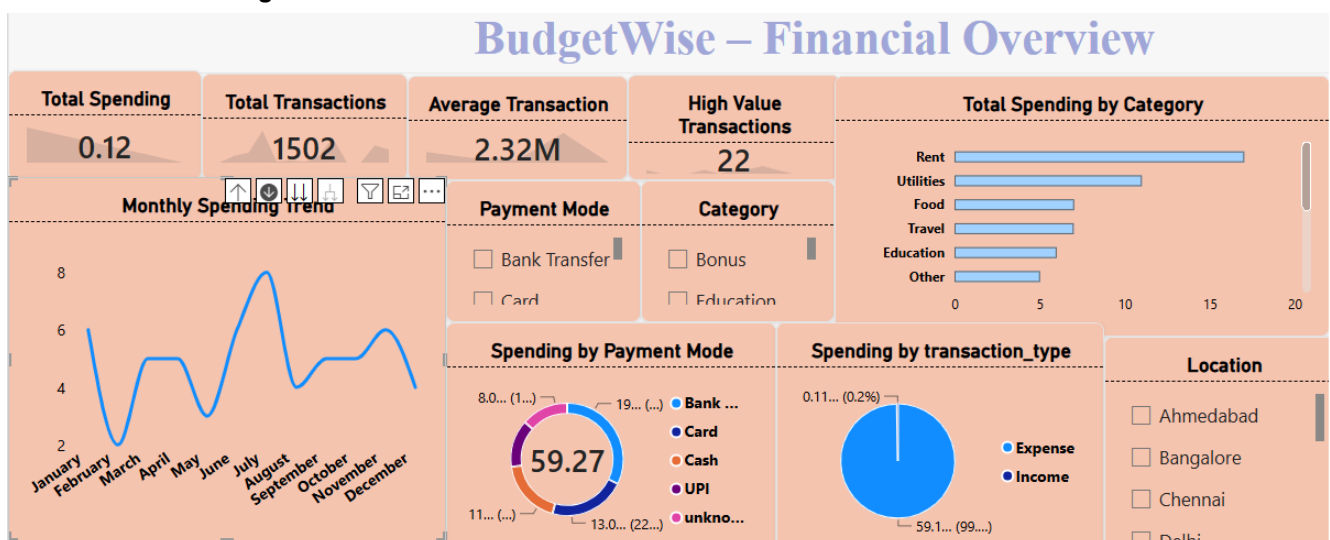
7. Data Modelling and DAX

Key DAX Measures:

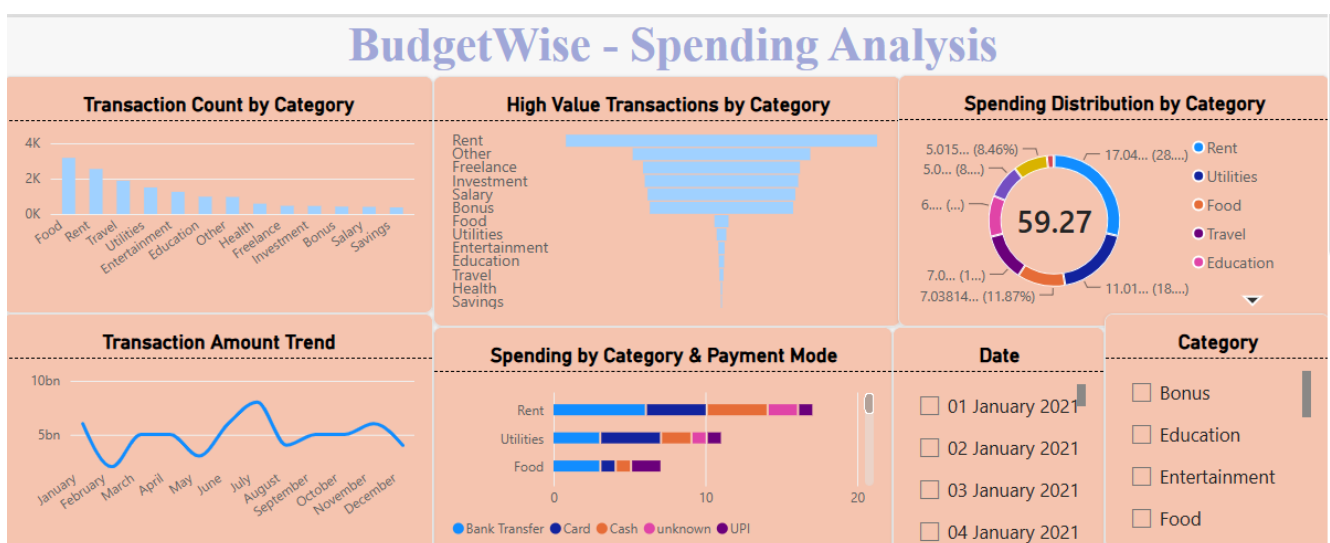
- Total Spending
- Total Transactions
- Average Transaction
- Maximum Transaction
- High Value Transactions

8. Power BI Analysis and Visualizations

Dashboard 1 – BudgetWise – Financial Overview



Dashboard 2 – BudgetWise – Spending Analysis



9. Insights & Conclusions

Key Findings

- Identified categories with higher transaction frequency.
- Compared total spending across categories.
- Analyzed spending by payment mode and transaction type.
- Identified high-value transaction patterns.
- Observed changes in transaction amounts over time.

Descriptive Analysis

- Summarized transaction counts, spending amounts, categories, payment methods, and transaction types.
- Identified spending distribution and frequently used categories.

2. Diagnostic Analysis

- Compared spending across categories, payment modes, and transaction types.
- Examined high-value transactions and spending changes over time to understand spending behaviour.

3. Predictive Analysis

- Predictive modelling and forecasting were not included in this project.
- The identified spending trends can provide a foundation for future predictive analysis.

4. Prescriptive Analysis

- Insights can support better budgeting and financial planning.
- Users can focus on high-spending categories, monitor high-value transactions, and improve spending management.

10. Conclusion

The BudgetWise identify major spending categories, transaction frequency, payment methods, high-value transactions, and spending trends over time. The interactive dashboards present these insights in a clear and user-friendly manner, enabling a better understanding of financial behaviour. Overall, the project provides data-driven insights that can support effective budgeting, improve spending awareness, and encourage better financial decision-making.